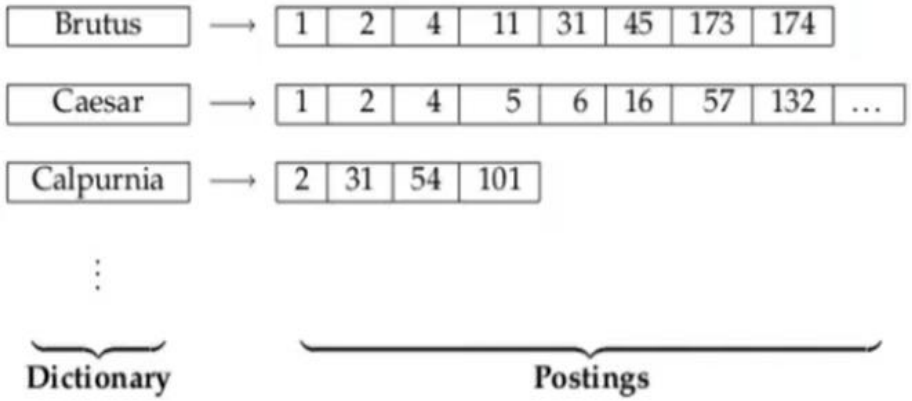
Intersect algorithm

Exercise 1

Exercise 0.1



► **Figure 1.3** The two parts of an inverted index. The dictionary is commonly kept in memory, with pointers to each postings list, which is stored on disk.

Draw the inverted index that would be built for the following document collection.
(See Figure 1.3 for an example.)
Doc 1: new home sales top forecasts
Doc 2: home sales rise in july
Doc 3: increase in home sales in july
Doc 4: july new home sales rise

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First step: tekonizer

Words	Documents
new	1
home	1
sales	1
top	1
forecasts	1
home	2
sales	2
rise	2
in	2
july	2
increase	3
in	3
home	3
sales	3
in	3
july	3
july	4
new	4
home	4
sales	4
rise	4

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Solution

First step: tekonizer

Words	Documents
new	1
home	1
sales	1
top	1
forecasts	1
home	2
sales	2
rise	2
in	2
july	2
increase	3
in	3
home	3
sales	3
in	3
july	3
july	4
new	4
home	4
sales	4
rise	4

second step: Sorting

Words	Documents
forecasts	1
home	1
home	2
home	3
home	4
in	2
in	3
in	3
increase	3
july	2
july	3
july	4
new	1
new	4
rise	2
rise	4
sales	1
sales	2
sales	3
sales	4
top	1

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Solution

First step: tekonizer

Words	Documents
new	1
home	1
sales	1
top	1
forecasts	1
home	2
sales	2
rise	2
in	2
july	2
increase	3
in	3
home	3
sales	3
in	3
july	3
july	4
new	4
home	4
sales	4
rise	4

second step: Sorting

Words	Documents
forecasts	1
home	1
home	2
home	3
home	4
in	2
in	3
in	3
increase	3
july	2
july	3
july	4
new	1
new	4
rise	2
rise	4
sales	1
sales	2
sales	3
sales	4
top	1

Dictionary	Posting List
forecasts	1
home	1,2,3,4
in	2,3
increase	3
july	2,3,4
new	1,4
rise	2,4
sales	1,2,3,4
top	1

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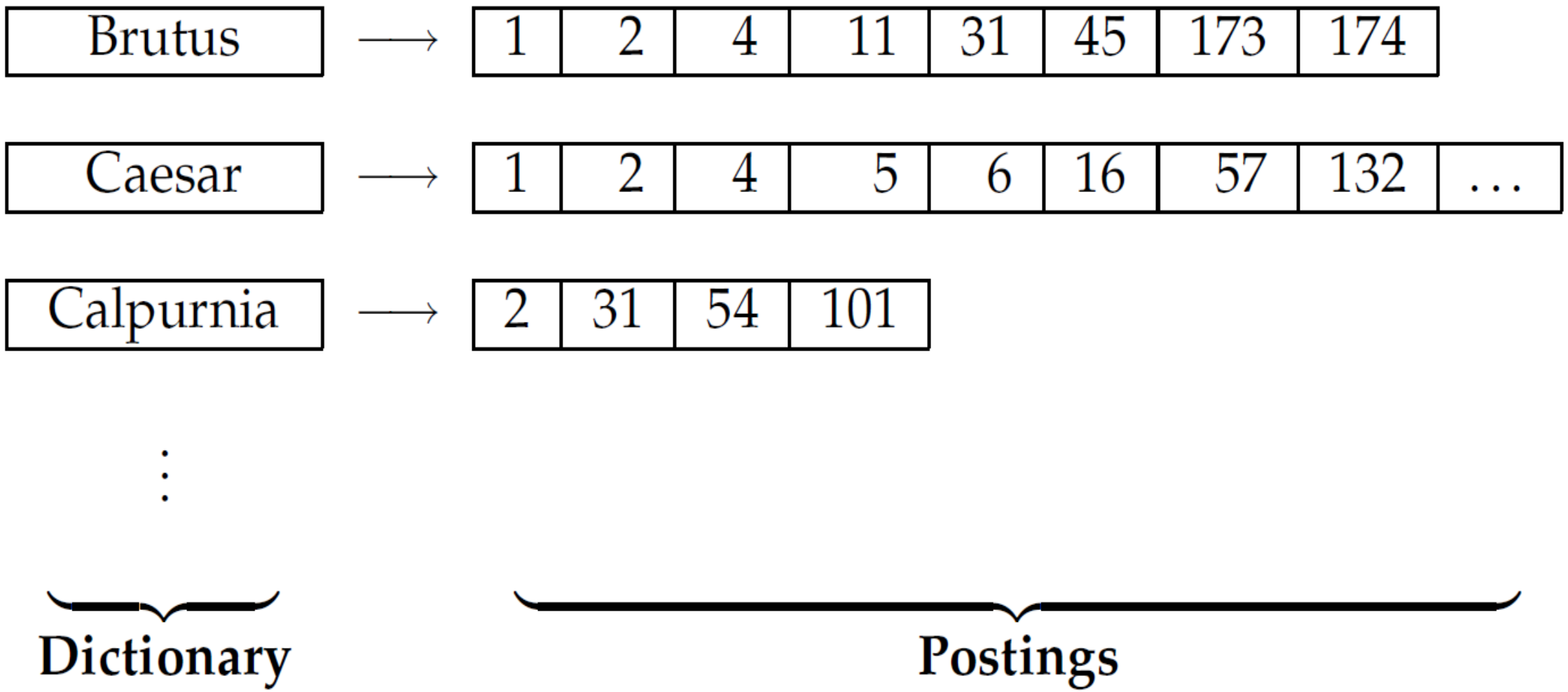

▶ **Figure 1.3** The two parts of an inverted index. The dictionary is commonly kept in memory, with pointers to each postings list, which is stored on disk.

Examples

1. Brutus AND Caesar AND Calprunia
2. Brutus OR Caesar OR Calprunia
3. Brutus AND Caesar OR Calprunia
4. Caesar OR Calprunia
5. Caesar AND NOT Calprunia
6. Brutus OR NOT Calprunia

Term Document

Words	Documents
approach	3
breakthrough	1
drug	1
drug	2
for	1
for	3
for	4
hopes	4
new	2
new	3
new	4
of	3
patients	4
schizophrenia	1
schizophrenia	2
schizophrenia	3
schizophrenia	4
treatment	3

Words	Doc1	Doc2	Doc3	Doc4
approach	0	0	1	0
breakthrou	1	0	0	0
drug	1	1	0	0
for	1	0	1	1
hopes	0	0	0	1
new	0	1	1	1
of	0	0	1	0
patients	0	0	0	1
schizophre	1	1	1	1
treatment	0	0	1	0

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Exercise 5

Exercise 0.22

Shown below is a portion of a positional index in the format:
term: doc1: (position1, position2, . . .) ; doc2: (position1, position2, . . .) ; etc.

angels :	2: (36,174,252,651)	4: (12,22,102,432)	7: (17)
fools :	2: (1,17,74,222)	4: (8,78,108,458)	7: (3,13,23,193)
fear :	2: (87,704,722,901)	4: (13,43,113,433)	7: (18,328,528)
in :	2: (3,37,76,444,851)	4: (10,20,110,470,500)	7: (5,15,25,195)
rush :	2: (2,66,194,321,702)	4: (9,69,149,429,569)	7: (4,14,404)
to :	2: (47,86,234,999)	4: (14,24,774,944)	7: (199,319,599,709)
tread :	2: (57,94,333)	4: (15,35,155)	7: (20,320)
where :	2: (67,124,393,1001)	4: (11,41,101,421,431)	7: (16,36,736)

Which document(s) if any meet each of the following queries, where each expression within quotes is a phrase query?

a. “fools rush in”
b. “fools rush in” AND “angels fear to tread”

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a. "fools rush in"
b. "fools rush in" AND "angels fear to tread"

Answer (a): All three documents (2, 4, and 7) satisfy the query.

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Which document(s) if any meet each of the following queries, where each expression within quotes is a phrase query?

a. "fools rush in"
b. "fools rush in" AND "angels fear to tread"

Answer (a): All three documents (2, 4, and 7) satisfy the query.

Answer (b): Only document 4.

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